

Lab 1: Requirements Engineering & UML Use-Case Modelling

Food Bank Surplus Redistribution Platform

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SECTION: D

Requirements Table

1.Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow donor partners to post surplus food packages with quantity, food type, dietary tags, pickup location, and expiry date.	High	Pass: Food batch is posted with all required details and an expiry countdown. Fail: Required information is missing.	Helps shelters find suitable surplus food before it expires.
FR-002	Functional	The system shall allow shelter coordinators to view and reserve available food batches and select suitable pickup windows.	High	Pass: Shelter successfully reserves an available batch and pickup window. Fail: Already reserved or expired batch can be reserved.	Ensures surplus food is allocated to shelters in an organized way.
FR-003	Functional	The system shall send notifications to verified shelters when relevant high-quantity or perishable food batches are posted nearby.	High	Pass: Eligible shelters receive a notification for a matching food batch. Fail: Unrelated or unverified shelters are notified.	Allows shelters to respond quickly to time-sensitive food donations.
FR-004	Functional	The system shall verify donor partners and shelter coordinators before allowing them to post, reserve, or manage food redistribution activities.	Medium	Pass: Only verified users can perform restricted actions. Fail: Unverified users can post or reserve food.	Improves trust and prevents misuse of the platform.
FR-005	Functional	The system shall generate pickup verification codes and maintain records of food reservations, pickups, and completed redistributions.	Medium	Pass: A valid verification code confirms a pickup and the transaction is recorded. Fail: Invalid code is accepted.	Provides traceability of donated food and completed pickups.

2.Non-Functional Requirements

ID	Type	Description	Priority	Acceptance Criteria	Rationale
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NFR-001	Performance & Security	The platform shall dispatch notifications to eligible shelters within 30 seconds when a high-quantity perishable food batch is listed.	High	Pass: Benchmarking confirms notifications are dispatched within the target time under simulated peak load. Fail: Notification delay exceeds the target.	Enables shelters to respond before perishable food expires.
NFR-002	Reliability & Availability	The platform shall maintain accurate reservation and expiry information and remain available during normal and peak food redistribution activity.	High	Pass: Reservations are not duplicated, expired batches are inactive, and the system remains available during peak load. Fail: Duplicate reservations or incorrect expiry status occurs.	Prevents food allocation errors and supports reliable redistribution.

3.Use-Case Flow Specification

Use Case: Reserve Surplus Food Batch

Primary Actor	Shelter Coordinator
Goal	To reserve an available surplus food batch and a pickup window before the food expires.
Preconditions	1. The shelter coordinator is verified and authenticated. 2. A donor partner has posted an active surplus food batch. 3. The food batch has not expired and is available for reservation.
Postconditions	1. The food batch is reserved for the shelter. 2. A pickup window is recorded. 3. A pickup verification code is generated and the reservation is stored.

Main Success Scenario

1. The Shelter Coordinator logs into the platform.
2. The system displays available surplus food batches.
3. The Shelter Coordinator selects a suitable food batch based on food type, quantity, dietary tags, location, and expiry.
4. The system checks that the selected batch is still available and has not expired.
5. The Shelter Coordinator selects an available pickup window.
6. The system reserves the food batch and records the pickup window.
7. The system generates a pickup verification code and confirms the reservation.

Alternate Flow

A1. Food Batch Expired or Already Reserved

1. The system checks the selected batch before reservation.
2. If the batch has expired or has already been reserved, the system does not allow the reservation.
3. The system informs the Shelter Coordinator and displays other available batches.

A2. No Suitable Pickup Window

1. The system checks available pickup windows.
2. If no suitable window is available, the system informs the Shelter Coordinator.
3. The coordinator can choose another food batch or try again later.

4.UML Use-Case Diagram

